

MAT1575, Classwork7, Fall2026

3.1 Integration by Parts (p.231-239)

1. Evaluate each of the following indefinite integrals:

(a) $\int x \sin(x^2) dx$ $u=x^2, du=2x dx, \frac{du}{2}=x dx$ $(b) \int x \sin(x) dx$? Integration by parts

$$= \int \sin(u) \frac{du}{2} = \frac{1}{2} \int \sin(u) du = -\frac{1}{2} \cos(x^2) + C$$

2. Review: **Product Rule for Differentiation.** Let f, g be differentiable functions. Then we have $(fg)' = f'g + fg'$

3. Theorem. **Integration By Parts.**

Let $u = f(x)$ and $v = g(x)$ be functions with continuous derivatives. Then, the integration-by-parts formula for the Integral involving these two functions is:

Proof: Let $h(x) = f(x)g(x)$. By product rule, we have $h'(x) = f'(x)g(x) + f(x)g'(x)$. Then

$$f(x)g(x) = h(x) = \int h'(x) dx = \int (f'(x)g(x) + f(x)g'(x)) dx = \int f'(x)g(x) dx + \int f(x)g'(x) dx$$

If $\int f(x)g'(x) dx$ is what we want to find, then $\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx$

4. Use Integration by parts to evaluate $\int x \sin x dx$ with

(a) $u = x$ and $dv = \sin x dx$
 $du = 1 dx$ $v = -\cos(x)$

(b) $u = \sin x$ and $dv = x dx$
 $du = \cos(x) dx$ $v = \frac{x^2}{2}$

$$\begin{aligned} \int x \sin(x) dx &= x \cdot (-\cos(x)) - \int (-\cos(x)) dx \\ &= -x \cos(x) + \int \cos(x) dx \\ &= -x \cos(x) + \sin(x) + C \end{aligned}$$

$$\begin{aligned} &= \sin(x) \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cos(x) dx \\ &= \dots \text{Can't do it.} \end{aligned}$$

5. How to choose u and dv ?

The acronym **LIATE** or **ILATE** can often help to take some of the guesswork out of our choices:

u					
	Logarithmic,	Inverse Trigonometric,	Algebraic,	Trigonometric,	Exponential
	dv				

The type of function in the integral that appear firsts in the list should be the

Note: The part chooses to be dv must be something we can integrate first choice of u

6. Evaluate the given indefinite integrals using Integration by Parts:

(a) $\int x e^{2x} dx$ $u=x, dv=e^{2x} dx, du=dx, v=\frac{e^{2x}}{2}$

$$\begin{aligned} &= x \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} dx \\ &= \frac{x e^{2x}}{2} - \frac{1}{2} \frac{e^{2x}}{2} + C \end{aligned}$$

(b) $\int \frac{\ln x}{x^3} dx$ $u=\ln x, dv=\frac{dx}{x^3}, du=\frac{dx}{x}, v=-\frac{1}{2x^2}$

$$\begin{aligned} &= \ln x \cdot \left(-\frac{1}{2x^2}\right) - \int \frac{1}{-2x^2} \cdot \frac{dx}{x} \\ &= \frac{\ln x}{-2x^2} + \frac{1}{2} \int \frac{dx}{x^3} \\ &= \frac{\ln x}{-2x^2} + \frac{1}{2} \left(-\frac{1}{2x^2}\right) + C \\ &= \frac{\ln x}{-2x^2} - \frac{1}{4x^2} + C \end{aligned}$$

(c) $\int \ln x dx$ $u=\ln(x), dv=dx, du=\frac{dx}{x}, v=x$

$$\begin{aligned} &= x \ln(x) - \int x \cdot \frac{dx}{x} \\ &= x \ln(x) - x + C \end{aligned}$$

7. Special Cases: Applying Integration by Parts When LIATE **Doesn't Work Directly.**

Evaluate $\int t^3 e^{t^2} dt$.

$$= t^2 \cdot \frac{e^{t^2}}{2} - \int \frac{e^{t^2}}{2} \cdot 2t dt = \frac{t^2 e^{t^2}}{2} - \int t e^{t^2} dt$$

$$= \frac{t^2 e^{t^2}}{2} - \frac{e^{t^2}}{2} + C$$

8. Special Cases: Applying Integration by Parts **More Than Once.**

Evaluate $\int x^2 e^{3x} dx$.

$$= x^2 \frac{e^{3x}}{3} - \int \frac{e^{3x}}{3} \cdot 2x dx = \frac{x^2 e^{3x}}{3} - \frac{2}{3} \int x e^{3x} dx$$

$$= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left(x \frac{e^{3x}}{3} - \int \frac{e^{3x}}{3} dx \right) = \frac{x^2 e^{3x}}{3} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} + C$$

9. Special Cases: Applying Integration by Parts **More Than Once.** (similar to the homework question 19.)

Evaluate $\int e^x \cos(x) dx$.

$$= e^x \cos(x) - \int e^x (-\sin(x)) dx = e^x \cos(x) + \int e^x \sin(x) dx$$

$$= e^x \cos(x) + e^x \sin(x) - \int e^x \cos(x) dx$$

$$2 \int e^x \cos(x) dx = e^x \cos(x) + e^x \sin(x) \Rightarrow \int e^x \cos(x) dx = \frac{e^x \cos(x) + e^x \sin(x)}{2} + C$$

10. Theorem: **Integration by Parts for Definite Integrals**

Let $u = f(x)$ and $v = g(x)$ be functions with continuous derivatives on $[a, b]$. Then

$$\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$$

Note: Here $uv \Big|_a^b = u(b) \cdot v(b) - u(a) \cdot v(a)$

11. Evaluate the given definite integrals:

(a) $\int_0^{\frac{\pi}{2}} \theta \cos(\theta) d\theta$

$$= \left[\theta \sin(\theta) \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin(\theta) d\theta = \left[\frac{\pi}{2} \cdot \sin\left(\frac{\pi}{2}\right) - 0 \cdot \sin(0) \right] - \left[-\cos(\theta) \right]_0^{\frac{\pi}{2}}$$

$$= \left[\frac{\pi}{2} - 0 \right] + \left[\cos\left(\frac{\pi}{2}\right) - \cos(0) \right] = \frac{\pi}{2} - 1$$

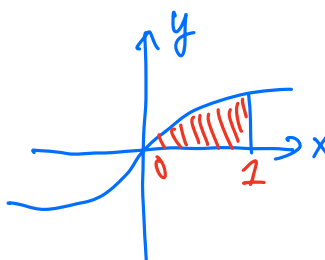
(b) $\int_1^2 x^3 \ln x dx$

$$= \left[\frac{x^4}{4} \ln(x) \right]_1^2 - \int_1^2 \frac{x^4}{4} \cdot \frac{1}{x} dx = \left[\frac{x^4}{4} \ln(x) \right]_1^2 - \int_1^2 \frac{x^3}{4} dx$$

$$= \left[\frac{2^4}{4} \ln(2) - \frac{1^4}{4} \ln(1) \right] - \left[\frac{x^4}{16} \right]_1^2$$

$$= 4 \ln(2) - \left[\frac{2^4}{16} - \frac{1^4}{16} \right] = 4 \ln(2) - \frac{15}{16}$$

12. Find the area of the region bounded above by the graph of $y = \tan^{-1} x$ and below by the x -axis over the interval $[0, 1]$



$\int_0^1 \tan^{-1}(x) dx = \left[x \tan^{-1}(x) \right]_0^1 - \int_0^1 x \frac{1}{x^2+1} dx$

$$= \left[1 \cdot \tan^{-1}(1) - 0 \cdot \tan^{-1}(0) \right] - \int_0^1 \frac{1}{x^2+1} dx$$

$$= \left[1 \cdot \frac{\pi}{4} - 0 \right] - \left[\frac{\ln(u)}{2} \right]_1^2$$

$$= \frac{\pi}{4} - \left[\frac{\ln(2)}{2} - \frac{\ln(1)}{2} \right] = \frac{\pi}{4} - \frac{\ln(2)}{2}$$

Let $\theta = \tan^{-1}(1) \Rightarrow \tan(\theta) = 1$
 $\Rightarrow \frac{\sin(\theta)}{\cos(\theta)} = 1 \Rightarrow \theta = \frac{\pi}{4}$

